AMPHIREACH AR-Mini – HARDWARE CIRCUIT DIAGRAM

SENSORS



MLX90642-D45
thermal · forward 45°

3V3
GND
SDA0
SCL0



MLX90640-110°
thermal · cradle

3V3
GND
SDA1
SCL1



BNO055
9-DoF IMU

3V3
GND
SDA0
SCL0

LEAK PROBE
2 bare wires · hull floor

OUT
GND

VBAT SENSE
100k / 33k divider

ADC
GND

MAIN CONTROLLER

(XIAO ESP32S3 SENSE)



XIAO ESP32S3 SENSE
OV2640 · microSD · WiFi

5V
GND
3V3
A0 VBAT
A1 LEAK/KILL
D2 SDA1
D3 SCL1
D4 SDA0
D5 SCL0

D6 TX1
D7 RX1
USB-C
D8–D10 SPI→SD
GPIO21 SD CS

USB-C ↔ USB-A

FreeRTOS · core 0: perception (thermal blobs, pan loop) · core 1: control (state machine, motors, CSV log)

PWM DRIVER

(PCA9685 · I2C 0x40 · 16 ch)

3V3
GND
SDA
SCL
OE



PCA9685
16-ch 12-bit PWM

CH0
CH1
CH2
CH3
CH4
CH5
CH6
CH7–9

KILL → OE: all PWM outputs off, in hardware

THRUSTERS



THRUSTER L · CCW



THRUSTER R · CW

→ prop

TRACK DRIVERS

(BTS7960 ×2 · 43 A)



BTS7960 L

M+ M−



BTS7960 R

M+ M− → ROB0154 · 12 V · 160 rpm

INDICATORS & SAFETY

BUZZER
NPN BC337

LED ×3
state · arm · fault



KILL SW

5 V → BTS7960 VCC · R_EN · L_EN

SERVO BUS

(Waveshare Serial Bus Servo Driver · UART 1 Mbps)



BUS DRIVER
TTL↔servo bus

TX
RX
12V
GND

5264-3P daisy chain
30 cm + 90 cm cables

ST3215 ID1
cradle L

ST3215 ID2
cradle R

ST3215 ID3
cam pan

ST3215 ID4
dome v2

RASPBERRY PI 5 (v2: ROS 2 bridge · telemetry)

USB-C PWR
USB-A



RASPBERRY PI 5
8 GB · active cooler · 64 GB



BUCK 12→5 V 5 A
USB-C out · 25 W



ROB0154 chassis

GND
XIAO ↔ Pi over USB (serial) · ROS 2 bags · WiFi dashboard

POWER SUPPLY



LiPo 3S 2200 mAh
XT60 · 40C



KILL SW 50 A



FUSE BLOCK
6-slot blade

20 A thrusters
15 A servo bus
10 A tracks
5 A logic
5 A Pi

12 V in



BUCK 12→5 V
3 A · logic rail

GND common

LEGEND

+12 V
+5 V
3V3
GND
PWM / signal
I2C
UART / bus
Motor out

Power circuits (thrusters, servo bus, tracks) come directly from the fuse block; only signals pass through the XIAO and the PCA9685. Common ground for all modules. Kill: 50 A battery switch (hardware) + PCA9685 OE (logic) + ADC ladder to A1.